

ORGANISMS AND POPULATIONS

1. At organismic level which type of ecology exists
 - (1) Synecology
 - (2) Physiological ecology
 - (3) Behavioural ecology
 - (4) Systematic ecology
2. Formation of different kind of biomes depends on
 - (1) Light
 - (2) Temperature
 - (3) Precipitation
 - (4) Both 2 and 3
3. Regional and local variation within each biome lead to formation of –
 - (1) Climate
 - (2) Weather
 - (3) Habitat
 - (4) Niche
4. What is / are key elements that leads to so much variation in the physical and chemical conditions of different habitats?
 - (1) Temperature
 - (2) Water and light
 - (3) Soil
 - (4) All above
5. Temperature is the most ecologically relevant environmental factor. In which of the following habitats temperature can exceed 100° C ?
 - (1) Tropical desert
 - (2) Thermal springs
 - (3) Deep sea hydrothermal vents
 - (4) Both 2 and 3
6. Find out the correct match with reference to their habitat –
 - (1) Mango tree – Canada
 - (2) Snow leopards – Kerela forest
 - (3) Tuna fish – Temperate latitudes in oceans
 - (4) Lion – Gujarat
7. Temperature is one of the important abiotic factor. Significance of temperature on living beings can be realised through –
 - (1) Kinetics of enzymes
 - (2) Basal metabolism
 - (3) Physiological functions
 - (4) All the above
8. Next to temperature, water is the most important factor influencing the life of organism. Which among the following water characteristics is not an influencing character?
 - (1) pH
 - (2) Turbidity
 - (3) Colour
 - (4) Salinity
9. What is the salinity of hypersaline lagoons?
 - (1) 5 ppt
 - (2) 30–35 ppt
 - (3) More than 100 ppt
 - (4) Less than 50 ppt
10. Among the following algae that inhabit the sea, which is likely to be found in the deepest water?
 - (1) Red algae
 - (2) Brown algae
 - (3) Green algae
 - (4) Golden brown algae
11. Percolation and water holding capacity of soil does not depend on –
 - (1) Soil composition
 - (2) Biota
 - (3) Size of grains
 - (4) Aggregation
12. Which of the following is main reason for non occurrence of small size conformers ?
 - (1) Karyoplasmic index
 - (2) Area : volume ratio
 - (3) Basal metabolism
 - (4) All the above
13. Which of the following alternative used by zooplanktons to overcome partial stressful conditions ?
 - (1) Migration
 - (2) Diapause
 - (3) Hibernation
 - (4) Aestivation
14. Majority of plants belongs to which of the following category
 - (1) Regulators
 - (2) Conformers
 - (3) Partial regulators
 - (4) Eurytherms

- 15.** Shortening of ears, limbs and other extremities of mammals so that heat loss can be minimise, is associated with –
- (1) Allen's rule (2) Bergeman's rule
(3) Jordan's rule (4) Rensch's rule
- 16.** Altitude sickness can be seen at which specific height ?
- (1) < 3500 m (2) > 3500 m
(3) 5300 m (4) < 530 m
- 17.** Which of the following is not an adaptation for altitude sickness ?
- (1) Increase in red blood cell production
(2) Decrease in binding capacity of oxygen with haemoglobin
(3) Increased breathing rate
(4) Increased heart palpitations
- 18.** Behavioural response to cope with variations in the environment can be seen in
- (1) CAM plants
(2) Opuntia plant
(3) Desert lizards
(4) C₄ - plants
- 19.** Population ecology is an important area of ecology because it links ecology with
- (1) Population genetics (2) Evolution
(3) Physiognomy (4) Both 1 and 2
- 20.** The tiger counting in our national parks and tiger reserves is often based on
- (1) Pug marks (2) Manual counting
(3) Fecal pellets (4) Both 1 and 3
- 21.** Match the following
- | | |
|--|--------------------------|
| (a) Breeding once in life | (i) Mammals |
| (b) Breeding several times in life | (ii) Oysters |
| (c) Large number of small sized offsprings | (iii) Most of birds |
| (d) Small number of large sized offsprings | (iv) Pacific salmon fish |
- (1) a(iv), b(ii), c(iii), d(i) (2) a(iv), b(iii), c(ii), d(i)
(3) a(iii), b(iv), c(ii), d(i) (4) a(ii), b(iii), c(iv), d(i)
- 22.** Match the following given population interactions
- | | |
|-----------|-------------------|
| (a) + / + | (i) Predation |
| (b) – / – | (ii) Ammensalism |
| (c) + / – | (iii) Competition |
| (d) – / 0 | (iv) Mutualism |
- (1) a(i), b(ii), c(iii), d(iv) (2) a(i), b(iii), c(ii), d(iv)
(3) a(iv), b(iii), c(i), d(ii) (4) a(iv), b(iii), c(ii), d(i)
- 23.** The famous 'Australian havoc' is associated with which of the following invasive species :-
- (1) Nile perch (2) Prickly pear cactus
(3) Red fox (4) Rabbit
- 24.** In rocky intertidal communities removal of which of the following predator became the cause of destruction of 10 species of invertebrates?
- (1) Monarch butterfly
(2) Starfish predator
(3) *Paramecium aurelia*
(4) Abingdon tortoise
- 25.** Phytophagous insects show which of the following interaction :-
- (1) Predation
(2) Competition
(3) Mutualism
(4) Commensalism
- 26.** Which of the following cannot be used by prey for defence against predator :-
- (1) Cardiac glycosides (2) Strychnine
(3) Nectar (4) Quinine
- 27.** Regarding competition find out the wrong statement.
- (1) Unrelated species could compete for same resource
(2) Fitness of one species is lowered in presence of other species
(3) Abingdon tortoise become extinct due to competitor starfish
(4) *Balanus* leads to exclusion of *Chthamalus* from rock coasts of Scotland

- 28.** Which of the following mean was used by warblers to avoid competition and coexist
- (1) Difference in foraging activities
 - (2) Habitat fragmentation
 - (3) Competitive release
 - (4) All of these
- 29.** Which of the following is not an adaptation of parasites for assurance of parasite host interaction
- (1) Loss of sensory organs
 - (2) Presence of adhesive organs
 - (3) Loss of digestive system
 - (4) Low reproductive potential
- 30.** Which of the following match is incorrect for commensalism interaction?
- (1) Epiphytes on trees
 - (2) Egrets with grazing cattles
 - (3) Hermit crab and sea anemone
 - (4) Sea anemone and clown fish
- 31.** During interaction between sea anemone and clown fish, which get benefitted :-
- (1) Sea anemone only (2) Clown fish only
 - (3) Both (4) Neither 1 nor 2
- 32.** Which of the following is not an example of coevolution?
- (1) Orchid and Bee
 - (2) Opuntia and Cactophagous moth
 - (3) Yucca and Pronuba
 - (4) Wasp and Fig